

THE OFF-LINE ZEROS AS A PONTRYAGIN INDEX: A TAIL RELATIVE-FORM-BOUND REDUCTION, AND TWO COMPUTATIONS CLOSING THE FINITE-DEFECT ROUTE

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ABSTRACT. We study the localized Weil form $\mathfrak{t} = \mathfrak{a} - \mathfrak{p}$ (archimedean minus prime) as a quadratic form on a Kreĭn space, and track not its sign but its *negative index* κ , equal to the number of off-line zeros. The hierarchy $\kappa = 0 \text{ (RH)} \subset \kappa < \infty \subset \kappa = \infty$ suggests an infinite-dimensional analogue of finitization: *finite defect* ($\kappa < \infty$, a Pontryagin space Π_κ), where Kreĭn–Langer theory bounds the non-real spectrum. We reduce $\kappa < \infty$ to a single relative form bound on the prime tail (RFB_X) and then close the two natural mechanisms for it, by computation on the validated band-limited engine. First, the relative-form-subordination cutoff scales with the band, $X^*(d) \sim e^{2d}/10$, so no fixed prime head captures the negative squares. Second, the saturation $\lambda_{\max} = 1$ of the prime-vs-archimedean pencil is *approached*, not attained: the near-critical multiplicity grows linearly with the band and the spectral gap closes, so the near-null space is infinite-dimensional. Hence the off-line zero count is not finite-rank detectable; the criticality is self-similar across log-prime scale; and the question returns, with no finite shortcut, to the semiboundedness of $\mathfrak{t}$ on an infinite-dimensional near-null space. No claim about RH is made.

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1. INTRODUCTION

The Weil explicit formula realizes a quadratic functional $\mathfrak{t}(g) = \sum_{\rho} h_g(\gamma_{\rho})$ over the nontrivial zeros $\frac{1}{2} + i\gamma_{\rho}$, with h_g the entire extension of $|\hat{g}|^2$; on-line zeros contribute nonnegatively, off-line zeros contribute indefinite quartets, and $\text{RH} \iff \mathfrak{t} \geq 0$ (Weil's criterion). Realized on a Kreĭn space [1], $\mathfrak{t}$ carries a *negative index*

$$\kappa = \#\{\text{negative squares of } \mathfrak{t}\} = \#\{\text{off-line zeros}\}.$$

This paper asks whether κ is *finite* — a question strictly weaker than RH ($\kappa = 0$) and strictly stronger than any zero-density estimate — because $\kappa < \infty$ would place $\mathfrak{t}$ in a Pontryagin space Π_{κ} , where the Kreĭn–Langer theory [2] gives a self-adjoint operator with at most κ non-real eigenvalue pairs and a definitizing structure. Finite defect is the infinite-dimensional analogue of the finitization (a finite-rank model) that the function-field proof of RH supplies geometrically.

We reduce $\kappa < \infty$ to a concrete inequality (§2) and then close its two natural mechanisms by computation (§§3–4). The outcome is negative for finite defect and structurally informative: the off-line negative squares are not confined to any finite set of primes, and the criticality of $\mathfrak{t}$ is carried by an infinite-dimensional, self-similar near-null space.

2. THE REDUCTION TO A TAIL RELATIVE FORM BOUND

On band-limited test functions write

$$\mathfrak{t}[g] = \mathfrak{a}[g] - \mathfrak{p}[g], \quad \mathfrak{p}[g] = 2 \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} \text{Re}(g \star \tilde{g})(\log n),$$

with $\mathfrak{a}$ the archimedean form (gamma-factor multiplier $\Omega(r) = \text{Re} \psi(\frac{1}{4} + \frac{ir}{2}) - \log \pi$, positive at height). Split the prime form at a cutoff X , $\mathfrak{p} = \mathfrak{p}_{\leq X} + \mathfrak{p}_{>X}$; the head $\mathfrak{p}_{\leq X}$ is finite rank.

Proposition 2.1 (Reduction). *If for some X the tail satisfies the relative form bound*

$$(\text{RFB}_X) \quad |\mathfrak{p}_{>X}[g]| \leq a \mathfrak{a}[g] + C \|g\|^2, \quad a < 1,$$

then $\mathfrak{t}$ has finitely many negative squares, $\kappa < \infty$.

Proof. $\mathfrak{t} = \mathfrak{a} - \mathfrak{p}_{\leq X} - \mathfrak{p}_{>X} \geq (1 - a)\mathfrak{a} - \mathfrak{p}_{\leq X} - C$. The form $(1 - a)\mathfrak{a} - C$ is bounded below; subtracting the finite-rank form $\mathfrak{p}_{\leq X}$ adds at most rank $\mathfrak{p}_{\leq X} < \infty$ negative squares. Hence $\kappa \leq \text{rank } \mathfrak{p}_{\leq X} < \infty$. $\square$

The quantity controlling negative squares is the one-sided maximal generalized eigenvalue $\lambda_{\max}(X, d) = \sup_g \mathfrak{p}_{>X}[g]/\mathfrak{a}[g]$ on band d (primes $\leq e^{2d}$); $\mathfrak{t}$ stops producing negative squares from the tail once $\lambda_{\max}(X, d) \leq 1$. (RFB_X) for a *fixed* X as $d \rightarrow \infty$ is exactly what Proposition 2.1 needs.

Remark 2.2 (Why this is hard, in one line). $\sum_{n \leq N} \Lambda(n)n^{-1/2} \sim 2\sqrt{N}$ dominates $\mathfrak{a} \sim \log$, so (RFB_X) holds only through cancellation: the prime tail must exhibit square-root cancellation *in the relative-form sense*. This is the same $\sqrt{N}$ -vs- $\log N$ obstruction met at the foundation of this line of work.

3. COMPUTATION I: THE SUBORDINATION CUTOFF SCALES WITH THE BAND

On the validated band-limited engine (sinc basis at spacing π/d , height $T_0 = 10^3$ where $\mathfrak{a} \succ 0$), we computed $\lambda_{\max}(X, d)$.

Numerical observation 3.1. For $d = 4$ ($e^{2d} \approx 2981$) and $d = 5$ ($e^{2d} \approx 22026$): $\lambda_{\max}(1, d) = 1.00000$ (the full form saturates, reproducing the Carleson-saturation result $C \equiv 1$); and $\lambda_{\max}(X, d)$ first drops below 1 at

$$X^*(d) \approx \frac{1}{10} e^{2d} \quad (X^*(4) \approx 300, \quad X^*(5) \approx 2000).$$

Equivalently, in log-scale the subordinate tail is the fixed-width top slice $[2d - \log 10, 2d]$, of width ≈ 2.3 independent of d ; for any fixed X , $\lambda_{\max}(X, d)$ increases with d and recrosses above 1.

Corollary 3.2. *(RFB_X) fails for every fixed X in the limit $d \rightarrow \infty$: the cutoff making the tail subordinate diverges, $X^*(d) \rightarrow \infty$. The finite-rank-head mechanism of Proposition 2.1 does not establish $\kappa < \infty$.*

Removing small primes *raises* $\lambda_{\max}$ above 1 before it falls, so the saturation at exactly 1 is a global property of all primes, not a small-prime artifact; the offending directions recur at every scale.

4. COMPUTATION II: THE SATURATION IS APPROACHED, NOT ATTAINED

We next ask whether the saturation $\lambda_{\max} = 1$ of the full pencil $(\mathfrak{p}, \mathfrak{a})$ is carried by finitely many directions. We computed the full spectrum for increasing band, with basis dimension scaled to the band.

Numerical observation 4.1. At $T_0 = 10^3$, the near-critical multiplicity of $(\mathfrak{p}, \mathfrak{a})$ grows linearly with the band:

$$\#\{\lambda > 0.99\} = 3, 11, 19, 27 \quad \text{for } d = 3, 4, 5, 6,$$

(constant increment 8), the spectral gap $\lambda_{\max} - \lambda_2$ is $\leq 10^{-4}$ throughout, and the near-critical fraction rises from 10% to 55% of the spectrum.

Corollary 4.2. *The saturation is approached: a subspace of growing dimension (indeed a growing positive fraction) satisfies $\mathfrak{p} \approx \mathfrak{a}$, i.e. $\mathfrak{t} \approx 0$. The near-null space of $\mathfrak{t}$ is infinite-dimensional in the limit. Hence the finite-defect picture fails: there is no finite critical structure to which a Π_κ realization with $\kappa < \infty$ could attach.*

5. CONCLUSION

Corollaries 3.2 and 4.2 close, by computation, both natural routes to $\kappa < \infty$: the off-line negative squares are neither captured by removing a finite prime head, nor concentrated in a finite critical subspace. Two structural facts emerge. The criticality is *self-similar*: the linear growth of the near-null dimension with the band (a log-prime range, equally a height-scale) matches the $\log T$ growth of the zero density, and the subordination window of Computation I slides to infinity at a fixed log-width. And the off-line zero count is *not finite-rank detectable*: any control of κ must face the form on an infinite-dimensional near-null space. The problem therefore returns to the semiboundedness of $\mathfrak{t}$ — this line of work’s central inequality — now equipped with the fact that no finite-defect shortcut exists. We make no claim about RH; we have determined the shape of the obstruction in the Pontryagin framework and shown it admits no finite reduction.

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